

# PHYS 512

Quantum Algorithms

# Deutsch's Algorithm (Concept)

Consider a function  $f(x)$  with single input and single output and  $x \in \{0, 1\}$

$f(x)$  can of three different types

Identity function

$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ 1 & \text{if } x = 1 \end{cases}$$

Constant function

$$f(x) = 0, \quad f(x) = 1$$

Bit flip function

$$f(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x = 1 \end{cases}$$

The **identity** and **bit flip** functions are called **balanced** because the outputs are opposite for half the inputs. **So a function on a single bit can be constant or balanced.**

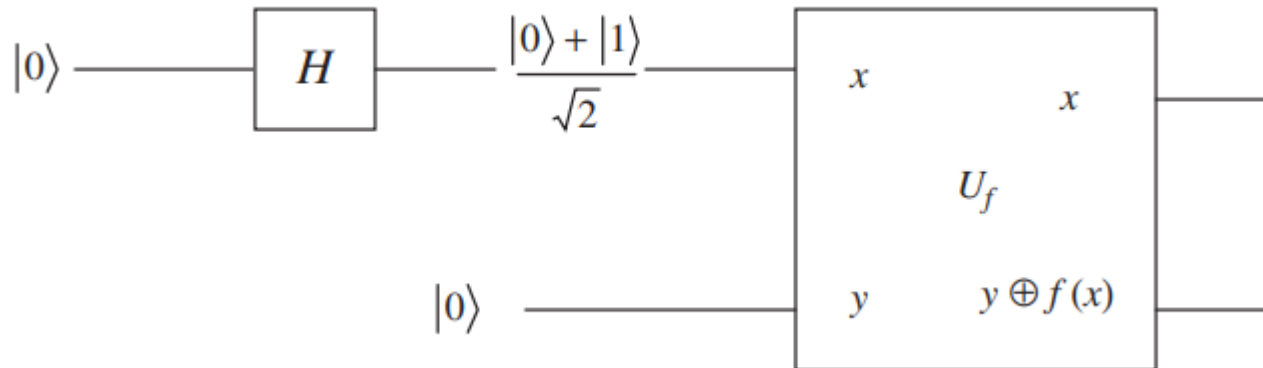
**Whether a function on a single bit is constant or balanced is a global property**

Deutsch's algorithm will let us put together a state that has all of the output values of the function associated with each input value in a superposition state. Then we will use quantum interference to find out if the given function is constant or balanced.

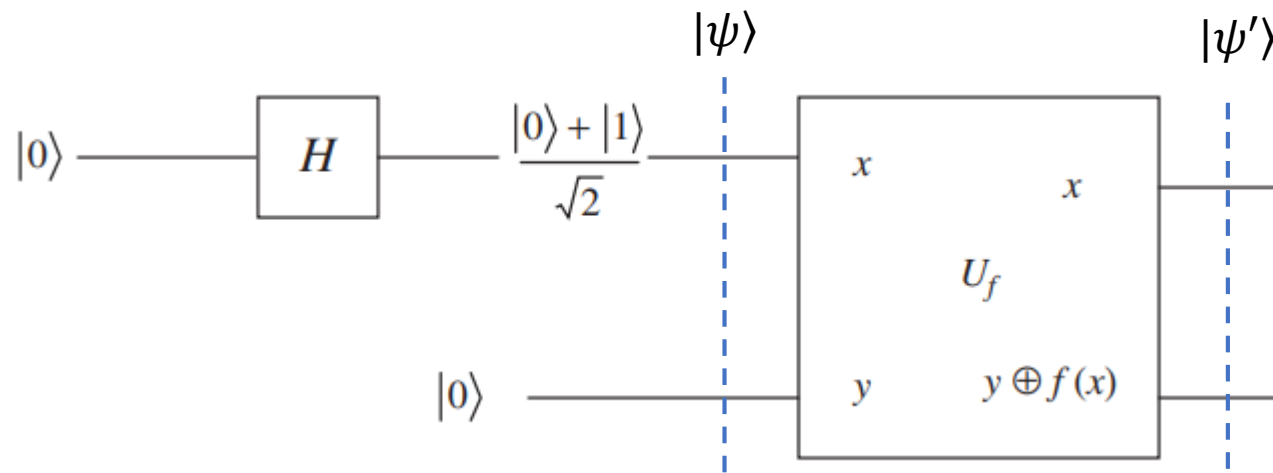
**Let us consider a unitary operator**

$$U_f|x, y\rangle = |x, y \oplus f(x)\rangle$$

Where  $x, y \in \{0, 1\}$



Circuit for the unitary  $U_f|x, y\rangle$



$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

$$U_f|x, y\rangle = |x, y \otimes f(x)\rangle$$

$$|\psi'\rangle = U_f|\psi\rangle = U_f \left[ \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \right]$$

$$|\psi'\rangle = \frac{1}{\sqrt{2}}(|0, 0 \otimes f(0)\rangle + |1, 0 \otimes f(1)\rangle)$$

$$|\psi'\rangle = \frac{1}{\sqrt{2}}(|0,0 \otimes f(0)\rangle + |1,0 \otimes f(1)\rangle)$$

If  $f(x)$  is a identity function

Identity function

$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ 1 & \text{if } x = 1 \end{cases}$$

$$|\psi'\rangle = \frac{1}{\sqrt{2}}(|0,0 \otimes f(0)\rangle + |1,0 \otimes f(1)\rangle)$$



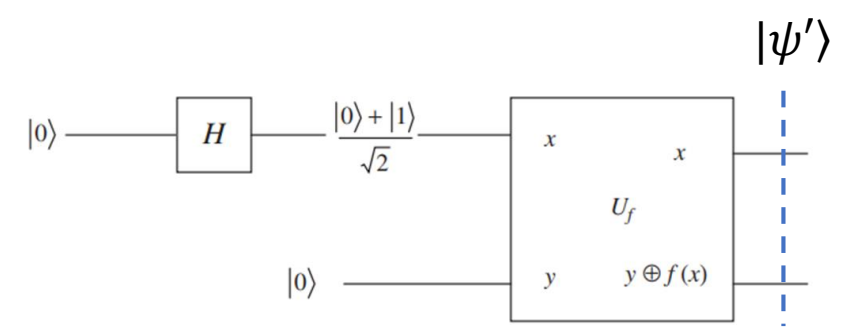
$$|\psi'\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

**In general**

$$|\psi'\rangle = \frac{1}{\sqrt{2}}(|0, f(0)\rangle + |1, f(1)\rangle)$$

$$|\psi'\rangle = \frac{1}{\sqrt{2}}(|0f(0)\rangle + |1f(1)\rangle)$$

Both function  $f(0)$  and  $f(1)$  are present in state  $|\psi'\rangle$

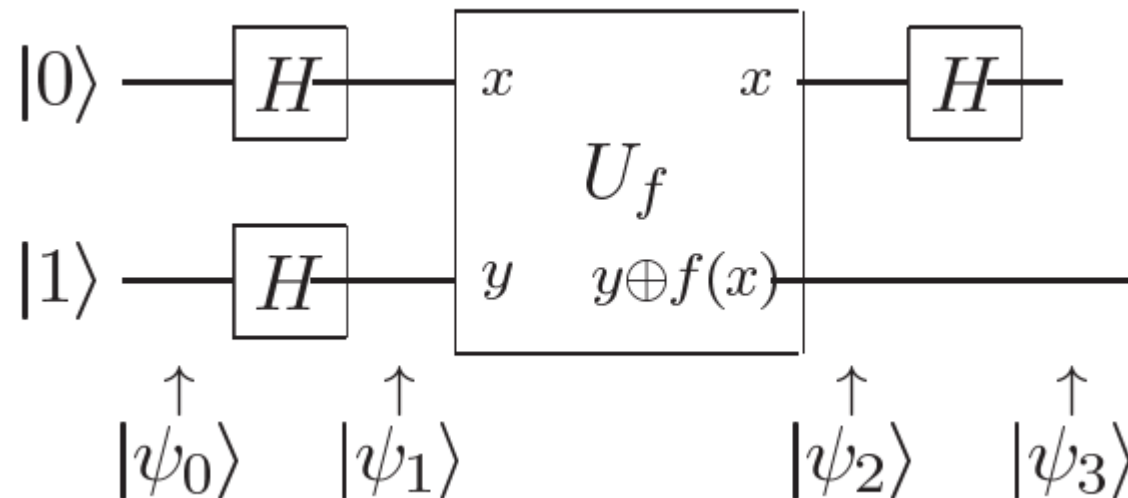


# Deutsch's algorithm

$$|\psi_{out}\rangle = (H \otimes I)U_f(H \otimes H)|0\rangle|1\rangle$$

In words Deutsch's algorithm is implemented by the following steps:

1. Apply Hadamard gates to the input state  $|0\rangle|1\rangle$  to produce a product state of two superpositions.
2. Apply  $U_f$  to that product state.
3. Apply a Hadamard gate to the first qubit leaving the second qubit alone.



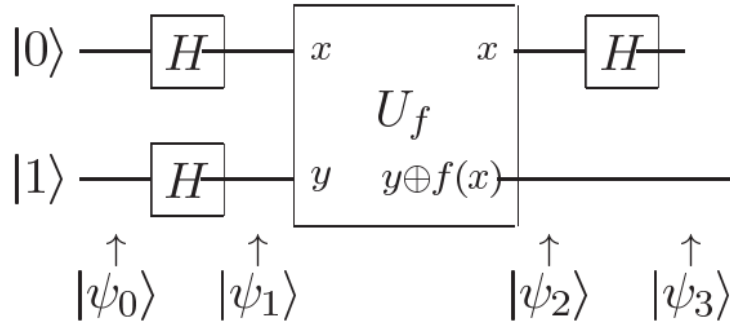
$$|\psi_0\rangle = |01\rangle$$

$$|\psi_1\rangle = HH|01\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$|\psi_1\rangle = HH|10\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

$$|\psi_2\rangle = U_f HH|10\rangle = \frac{1}{2}(|0,0\oplus f(0)\rangle - |0,1\oplus f(0)\rangle + |1,0\oplus f(1)\rangle - |1,1\oplus f(1)\rangle)$$

$$U_f|x,y\rangle = |x,y\oplus f(x)\rangle$$



$f(0) = f(1)$

$\Rightarrow$

$f(0) = 0$

$f(0) = 1$

$(|0,0\rangle - |0,1\rangle + |1,0\rangle - |1,1\rangle)$

$(|0,1\rangle - |0,0\rangle + |1,1\rangle - |1,0\rangle) = -(|0,0\rangle - |0,1\rangle + |1,0\rangle - |1,1\rangle)$

Therefore

$$|\psi_2\rangle = \pm(|0,0\rangle - |0,1\rangle + |1,0\rangle - |1,1\rangle) = \pm \left[ \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right]$$

$|\psi_2\rangle = \pm \left[ \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right]$

$f(0) = f(1)$

$$|\psi_2\rangle = \pm \left[ \left( \frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right] \quad \boxed{f(0) = f(1)}$$

Similarly for balanced function

$$|\psi_2\rangle = \pm \left[ \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right] \quad \boxed{f(0) \neq f(1)}$$

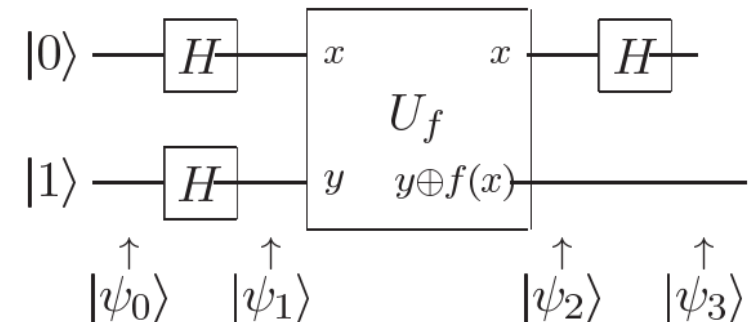
Remember  $f(x) = 0$  or  $1$

After applying H-gate to first qubit

$$|\psi_3\rangle = \pm |0\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \quad \boxed{f(0) = f(1)} \quad \text{Constant}$$

$$|\psi_3\rangle = \pm |1\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \quad \boxed{f(0) \neq f(1)} \quad \text{Balanced}$$

Therefore, by **measuring the first qubit**, we can determine whether the function is **constant or balanced**.

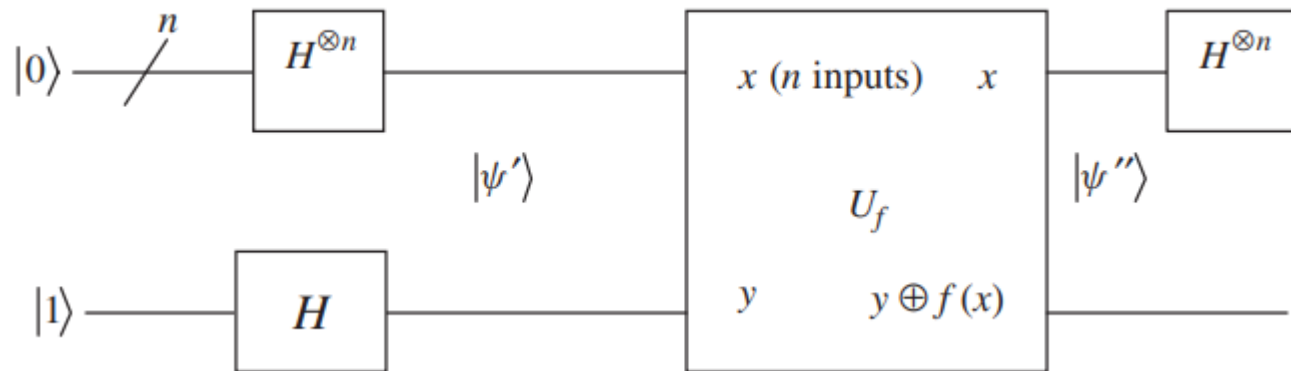




# Deutsch-Jozsa algorithm

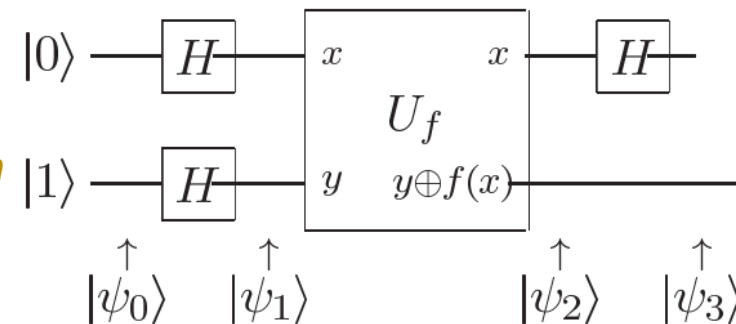
The **Deutsch-Jozsa algorithm** is a generalization of *Deutsch's algorithm*. Again, this algorithm allows us to determine whether a function  $f(x)$  is constant or balanced, but this time the function has multiple input values.

If  $f(x)$  is **constant**, then the output is the same for all input values  $x$ . If the function is **balanced**, then  $f(x) = 0$  for half of the inputs and  $f(x) = 1$  for the other half of the inputs, and vice versa



Deutsch-Jozsa algorithm

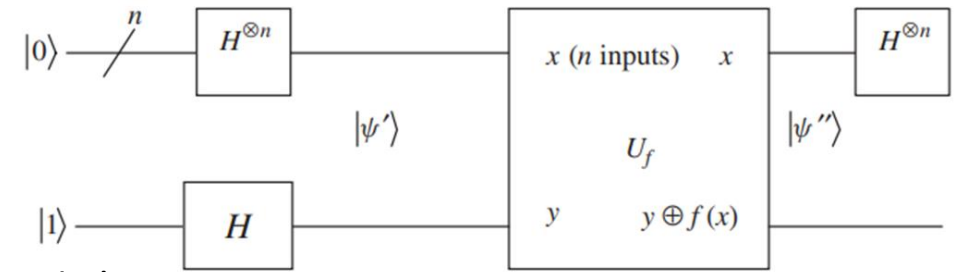
Deutsch's algorithm



$$|\psi'\rangle = (H^{\otimes n})(|0\rangle^{\otimes n}) \otimes (H|1\rangle)$$

$$|\psi'\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \text{ Where } |x\rangle \text{ is combinations of '0 and 1'}$$

$$|\psi''\rangle = \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$



Now apply H-gate to n-qubit string of 'first qubit' exp., consider

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_y (-1)^{x \cdot z} |z\rangle$$

To understand above

$$H|x\rangle = \sum_z (-1)^{xz} |z\rangle / \sqrt{2}. \quad \Rightarrow \quad H|x\rangle = \frac{1}{\sqrt{2}} [(-1)^{x_0} |0\rangle + (-1)^{x_1} |1\rangle]$$

$$H|0\rangle = \frac{1}{\sqrt{2}} [(-1)^{00} |0\rangle + (-1)^{01} |1\rangle] = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}} [(-1)^{10} |0\rangle + (-1)^{11} |1\rangle] = |-\rangle$$

$$H^{\otimes n}|x_1, \dots, x_n\rangle = \frac{\sum_{z_1, \dots, z_n} (-1)^{x_1 z_1 + \dots + x_n z_n} |z_1, \dots, z_n\rangle}{\sqrt{2^n}}$$

$$|\psi''\rangle = \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

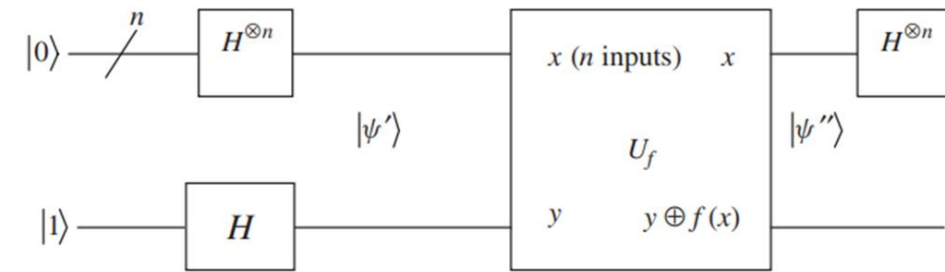
$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_y (-1)^{x \cdot z} |z\rangle$$

Final output from above two eqs.

$$|\psi_{out}\rangle = \frac{1}{2^n} \sum_z \sum_x (-1)^{x \cdot z + f(x)} |z\rangle \left( \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Now we measure the n inputs. **It might not be immediately obvious looking**  $|\psi_{out}\rangle$ , at but **there are two possible measurement results** on  $|z\rangle$  (which is the state of n inputs at this stage) that are of interest. The possible results are as follows

- Measurement of the **first n input qubits** in  $|\psi_{out}\rangle$  returns all 0's. In this case  $f(x)$  is **constant**.
- Otherwise, if at **least one of the qubits** in  $|z\rangle$  is found to be a 1 on measurement,  $f(x)$  is **balanced**



# Quantum Fourier Transformation

# Fourier Series (Background)

A function  $f(x)$  that is periodic with period  $L$  and square integrable in interval  $-L/2 < x < L/2$  allows the representation

$$f(x) = \sum_{n=0}^{\infty} a_n \cos\left(\frac{2\pi n x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{2\pi n x}{L}\right)$$

where the constants  $a_n, b_n$  are given by

$$a_0 = \frac{1}{L} \int_{-L/2}^{L/2} f(x) dx$$

$$a_n = \frac{2}{L} \int_{-L/2}^{L/2} f(x) \cos\left(\frac{2\pi n x}{L}\right) dx$$

$$b_n = \frac{2}{L} \int_{-L/2}^{L/2} f(x) \sin\left(\frac{2\pi n x}{L}\right) dx,$$

and  $n$  is a positive integer.

Today, Fourier analysis is found in almost all areas of science, technology, and the arts, from **quantum field theory to image processing**. Often, it is convenient to express Fourier series in terms of exponential functions. Using the Euler formula



# QFT Matrix - $F_n$

$$F_n = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \dots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \omega^6 & \dots & \omega^{2(n-1)} \\ 1 & \omega^3 & \omega^6 & \omega^9 & \dots & \omega^{3(n-1)} \\ 1 & \cdot & \cdot & \cdot & \dots & \cdot \\ 1 & & & & & \omega^{(n-1)(n-1)} \end{pmatrix}$$

Where  $\omega_n^k = e^{\frac{2\pi i k}{n}}$  and  $|\omega_n^k| = 1$

Show that  $\omega_n^k = \omega_n^{k+n} = \Rightarrow e^{\frac{2\pi i(k+n)}{n}} = e^{\frac{2\pi i k}{n}}$

For single qubit,  $n = 1$

$$F_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & \omega^2 \end{pmatrix} \quad \omega^2 = -1$$



$$F_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Which is Hadamard gate !

Apply it to single qubit state, like

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

For two qubits,  $n = 2$

$$F_4 = \frac{1}{\sqrt{4}} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & \omega^4 & \omega^6 \\ 1 & \omega^3 & \omega^6 & \omega^9 \end{pmatrix}$$



$$F_4 = \frac{1}{\sqrt{4}} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$$

Apply it to two qubit state, like

$$|11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

# QFT in Summation Form and Single Qubit QFT

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{i2\pi jk}{N}} |k\rangle$$

$$|j\rangle = |j_{n-1}\rangle \otimes |j_{n-2}\rangle \otimes |j_{n-3}\rangle \dots \otimes |j_1\rangle \otimes |j_0\rangle$$

$$|k\rangle = |k_{n-1}\rangle \otimes |k_{n-2}\rangle \otimes |k_{n-3}\rangle \dots \otimes |k_1\rangle \otimes |k_0\rangle$$

Integer/decimal of  
state  $|j\rangle$  and  $|k\rangle$

$$j = 2^{n-1}j_{n-1} + 2^{n-2}j_{n-2} + 2^{n-3}j_{n-3} + \dots 2^1j_1 + 2^0j_0$$

$$k = 2^{n-1}k_{n-1} + 2^{n-2}k_{n-2} + 2^{n-3}k_{n-3} + \dots 2^1k_1 + 2^0k_0$$

For single qubit QFT is simply  
Hadamard transformation

Example: single qubit  
 $n=1 \rightarrow N=2$

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{i2\pi jk}{N}} |k\rangle$$

$$\text{QFT}|0\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^1 e^{\frac{i2\pi(0)k}{2}} |k\rangle$$

$$|\tilde{0}\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^1 e^0 |k\rangle$$

$$|\tilde{0}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$\text{QFT}|1\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^1 e^{\frac{i2\pi(1)k}{2}} |k\rangle$$

$$|\tilde{0}\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^1 e^{i\pi k} |k\rangle$$

$$|\tilde{1}\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

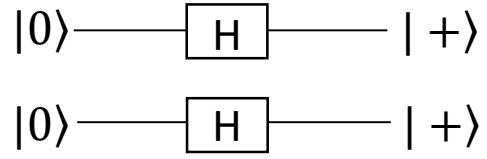


# 2 Qubits QFT

Example: Two qubit  
n=2 → N=4

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{i2\pi jk}{N}} |k\rangle$$

$$\text{QFT}|00\rangle = \frac{1}{\sqrt{2^2}} \sum_{k=0}^3 e^{\frac{i2\pi(0)k}{4}} |k\rangle \Rightarrow |\widetilde{00}\rangle = \frac{1}{\sqrt{2^2}} \sum_{k=0}^3 |k\rangle = \frac{1}{\sqrt{2^2}} (|0\rangle + |1\rangle + |2\rangle + |3\rangle)$$

$$|\widetilde{00}\rangle = \frac{1}{\sqrt{2^2}} (|00\rangle + |01\rangle + |10\rangle + |11\rangle) \Rightarrow |\widetilde{00}\rangle = \left[ \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right] \otimes \left[ \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right] \Rightarrow$$


$$|\widetilde{00}\rangle = |+\rangle \otimes |+\rangle$$

$$\text{QFT}|01\rangle = \frac{1}{\sqrt{2^2}} \sum_{k=0}^3 e^{\frac{i2\pi(1)k}{4}} |k\rangle \Rightarrow |\widetilde{01}\rangle = \frac{1}{\sqrt{2^2}} \sum_{k=0}^3 e^{\frac{i\pi k}{2}} |k\rangle = \frac{1}{\sqrt{2^2}} (e^0|0\rangle + e^{\frac{i\pi}{2}}|1\rangle + e^{i\pi}|2\rangle + e^{\frac{i3\pi}{2}}|3\rangle)$$

$$|\widetilde{01}\rangle = \frac{1}{\sqrt{2^2}} (|00\rangle + i|01\rangle - |10\rangle - i|11\rangle) \Rightarrow |\widetilde{01}\rangle = \left[ \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \right] \otimes \left[ \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle) \right] \Rightarrow |\widetilde{01}\rangle = |-\rangle \otimes |+i\rangle$$

# 2-Qubits Generalized QFT

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{i2\pi jk}{N}} |k\rangle$$

$$|j\rangle = |j_{n-1}\rangle \otimes |j_{n-2}\rangle \otimes |j_{n-3}\rangle \dots \otimes |j_1\rangle \otimes |j_0\rangle$$

Example: Two qubit  
n=2 → N=4

$$\text{QFT}|j\rangle = \text{QFT}|j_1j_0\rangle = \frac{1}{\sqrt{2^2}} \sum_{k=0}^3 e^{\frac{i2\pi jk}{4}} |k\rangle$$



$$|\widetilde{j_1j_0}\rangle = \frac{1}{\sqrt{2^2}} \sum_{k_1=0}^1 \sum_{k_0=0}^1 e^{\frac{i2\pi j(2k_1+k_0)}{4}} |k_1k_0\rangle$$

$$|\widetilde{j_1j_0}\rangle = \frac{1}{\sqrt{2}} \sum_{k_1=0}^1 e^{\frac{i2\pi j(2k_1)}{4}} |k_1\rangle \otimes \frac{1}{\sqrt{2}} \sum_{k_0=0}^1 e^{\frac{i2\pi j(k_0)}{4}} |k_0\rangle$$

$$|\widetilde{j_1j_0}\rangle = \frac{1}{\sqrt{2}} (e^0|0\rangle + e^{i\pi j}|1\rangle) \otimes (e^0|0\rangle + e^{\frac{i\pi j}{2}}|1\rangle)$$

$$|\widetilde{j_1j_0}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i2\pi j_1} e^{i\pi j_0}|1\rangle) \otimes (|0\rangle + e^{i\pi j_1} e^{\frac{i\pi j_0}{2}}|1\rangle)$$

$$|\widetilde{j_1j_0}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi j_0}|1\rangle) \otimes (|0\rangle + e^{i\pi j_1} e^{\frac{i\pi}{2}j_0}|1\rangle)$$

Now substitute  $j = 2j_1 + j_0$

$$e^{i2\pi j} = 1$$

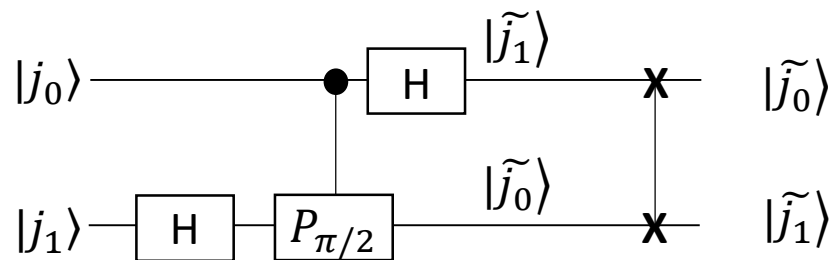
# 2-Qubits Generalized QFT

$$|j_1 \widetilde{j}_0\rangle = \underbrace{\frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi j_0} |1\rangle)}_{\text{Hadamard gate}} \otimes \underbrace{\frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi j_1} e^{\frac{i\pi}{2} j_0} |1\rangle)}_{\text{Hadamard} + \pi/2 \text{ rotation}}$$

For  $j_0 = 0 \rightarrow 1$   
 $j_0 = 1 \rightarrow -1$

$j_0$  dependent  
 $\frac{\pi}{2}$  rotation

First terms depends upon  $j_0$  NOT on  $j_1$



QFT Circuit for any combination of two qubits input states

# Quantum Fourier Transform

$$|\psi\rangle = \sum_{j=0}^{N-1} x_j |j\rangle \quad \text{Generalized quantum state}$$

Find QFT transformed state

$$QFT|\psi\rangle = \sum_{j=0}^{N-1} x_j QFT|j\rangle$$

$$QFT|\psi\rangle = \sum_{j=0}^{N-1} x_j \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{i2\pi jk}{N}} |k\rangle \langle j|j\rangle$$

$$QFT|\psi\rangle = \sum_{k=0}^{N-1} \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j e^{\frac{i2\pi jk}{N}} |k\rangle$$

$$|\tilde{\psi}\rangle = \sum_{k=0}^{N-1} y_k |k\rangle \quad \text{Where} \quad y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j e^{\frac{i2\pi jk}{N}}$$

QFT transforms quantum state bases

$$QFT|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{i2\pi jk}{N}} |k\rangle \quad \text{Transformed state}$$

$$QFT \text{ operator} = \hat{U}_Q = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{i2\pi jk}{N}} |k\rangle \langle j|$$

$\hat{U}_Q$  is a Unitary

# n-Qubits QFT

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{i2\pi jk}{N}} |k\rangle$$

$$|j\rangle = |j_{n-1}\rangle \otimes |j_{n-2}\rangle \otimes |j_{n-3}\rangle \dots \otimes |j_1\rangle \otimes |j_0\rangle$$

$$|k\rangle = |k_{n-1}\rangle \otimes |k_{n-2}\rangle \otimes |k_{n-3}\rangle \dots \otimes |k_1\rangle \otimes |k_0\rangle$$

QFT of n-qubit string

After expanding the k-summation

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{2^n}} \sum_{k_{n-1}=0}^1 \sum_{k_{n-2}=0}^1 \dots \sum_{k_1=0}^1 \sum_{k_0=0}^1 e^{\frac{i2\pi j}{2^n} (2^{n-1}k_{n-1} + 2^{n-2}k_{n-2} \dots 2^1k_1 + 2^0k_0)} |k_{n-1}\rangle \otimes |k_{n-2}\rangle \otimes \dots |k_1\rangle \otimes |k_0\rangle$$

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{2^n}} \sum_{k_{n-1}=0}^1 \sum_{k_{n-2}=0}^1 \dots \sum_{k_1=0}^1 \sum_{k_0=0}^1 e^{i\pi j \left( k_{n-1} + \frac{1}{2}k_{n-2} \dots \frac{1}{2^{n-2}}k_1 + \frac{1}{2^{n-1}}k_0 \right)} |k_{n-1}\rangle \otimes |k_{n-2}\rangle \otimes \dots |k_1\rangle \otimes |k_0\rangle$$

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{2}} \sum_{k_{n-1}=0}^1 e^{i\pi j(k_{n-1})} |k_{n-1}\rangle \otimes \frac{1}{\sqrt{2}} \sum_{k_{n-2}=0}^1 e^{i\pi j \frac{1}{2}(k_{n-2})} |k_{n-2}\rangle \otimes \dots \frac{1}{\sqrt{2}} \sum_{k_1=0}^1 e^{i\pi j \frac{1}{2^{n-2}}(k_1)} |k_1\rangle \otimes \frac{1}{\sqrt{2}} \sum_{k_0=0}^1 e^{i\pi j \frac{1}{2^{n-1}}(k_0)} |k_0\rangle$$

# n-Qubits QFT

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{2}} \sum_{k_{n-1}=0}^1 e^{i\pi j(k_{n-1})} |k_{n-1}\rangle \otimes \frac{1}{\sqrt{2}} \sum_{k_{n-2}=0}^1 e^{i\pi j \frac{1}{2}(k_{n-2})} |k_{n-2}\rangle \otimes \dots \frac{1}{\sqrt{2}} \sum_{k_1=0}^1 e^{i\pi j \frac{1}{2^{n-2}}(k_1)} |k_1\rangle \otimes \frac{1}{\sqrt{2}} \sum_{k_0=0}^1 e^{i\pi j \frac{1}{2^{n-1}}(k_0)} |k_0\rangle$$

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{2}} (e^0|0\rangle + e^{i\pi j}|1\rangle) \otimes \frac{1}{\sqrt{2}} (e^0|0\rangle + e^{i\pi j \frac{1}{2}}|1\rangle) \dots \otimes \frac{1}{\sqrt{2}} (e^0|0\rangle + e^{i\pi j \frac{1}{2^{n-2}}}|1\rangle) \otimes \frac{1}{\sqrt{2}} (e^0|0\rangle + e^{i\pi j \frac{1}{2^{n-1}}}|1\rangle)$$

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi j}|1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi j \frac{1}{2}}|1\rangle) \dots \otimes \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi j \frac{1}{2^{n-2}}}|1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi j \frac{1}{2^{n-1}}}|1\rangle)$$

Now expand j summation and consider terms one by one

$$|j\rangle = |j_{n-1}\rangle \otimes |j_{n-2}\rangle \otimes |j_{n-3}\rangle \dots \otimes |j_1\rangle \otimes |j_0\rangle$$

First Term

$$|\widetilde{j_{n-1}}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi(2^{n-1}j_{n-1} + 2^{n-2}j_{n-2} \dots 2^1j_1 + 2^0j_0)}|1\rangle)$$

$$e^{i2\pi j} = 1$$

$$|\widetilde{j_{n-1}}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi j_0}|1\rangle)$$

Hadamard gate

## Second term

$$|j_{n-2}^{\sim}\rangle = \frac{1}{\sqrt{2}} \left( |0\rangle + e^{i\pi\frac{1}{2}(2^{n-1}j_{n-1}+2^{n-2}j_{n-2}....2^1j_1+2^0j_0)} |1\rangle \right) \quad e^{i2\pi j} = 1$$

$$|j_{n-2}^{\sim}\rangle = \frac{1}{\sqrt{2}} \left( |0\rangle + e^{i\pi j_1} e^{i\frac{\pi}{2}j_0} |1\rangle \right)$$

**Hadamard + controlled  $\pi/2$  rotation**

## Second last term

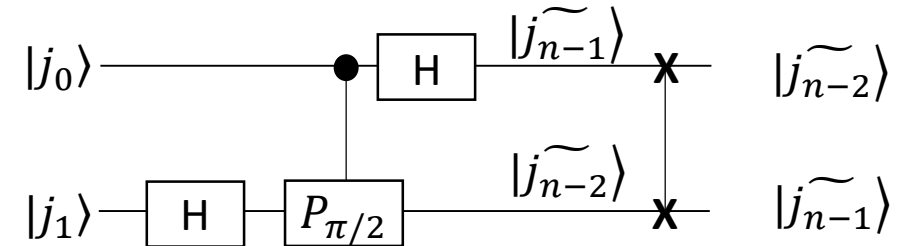
$$|\tilde{j}_1\rangle = \frac{1}{\sqrt{2}} \left( |0\rangle + e^{i\pi\frac{1}{2^{n-2}}(2^{n-1}j_{n-1}+2^{n-2}j_{n-2}....2^1j_1+2^0j_0)} |1\rangle \right)$$

$$|\tilde{j}_1\rangle = \frac{1}{\sqrt{2}} \left( |0\rangle + e^{i\pi j_{n-2}} e^{i\frac{\pi}{2}j_{n-3}} ..... e^{i\frac{\pi}{2^{n-3}}j_1} e^{i\frac{\pi}{2^{n-2}}j_0} |1\rangle \right)$$

## The last term

$$|\tilde{j}_0\rangle = \frac{1}{\sqrt{2}} \left( |0\rangle + e^{i\pi\frac{1}{2^{n-1}}(2^{n-1}j_{n-1}+2^{n-2}j_{n-2}....2^1j_1+2^0j_0)} |1\rangle \right)$$

$$|\tilde{j}_0\rangle = \frac{1}{\sqrt{2}} \left( |0\rangle + e^{i\pi j_{n-1}} e^{i\frac{\pi}{2}j_{n-2}} ..... e^{i\frac{\pi}{2^{n-2}}j_1} e^{i\frac{\pi}{2^{n-1}}j_0} |1\rangle \right)$$



# n-Qubits QFT - Circuit

$$|\widetilde{j_{n-1}}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi j_0} |1\rangle)$$

$$|\widetilde{j_{n-2}}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi j_1} e^{i\frac{\pi}{2} j_0} |1\rangle)$$

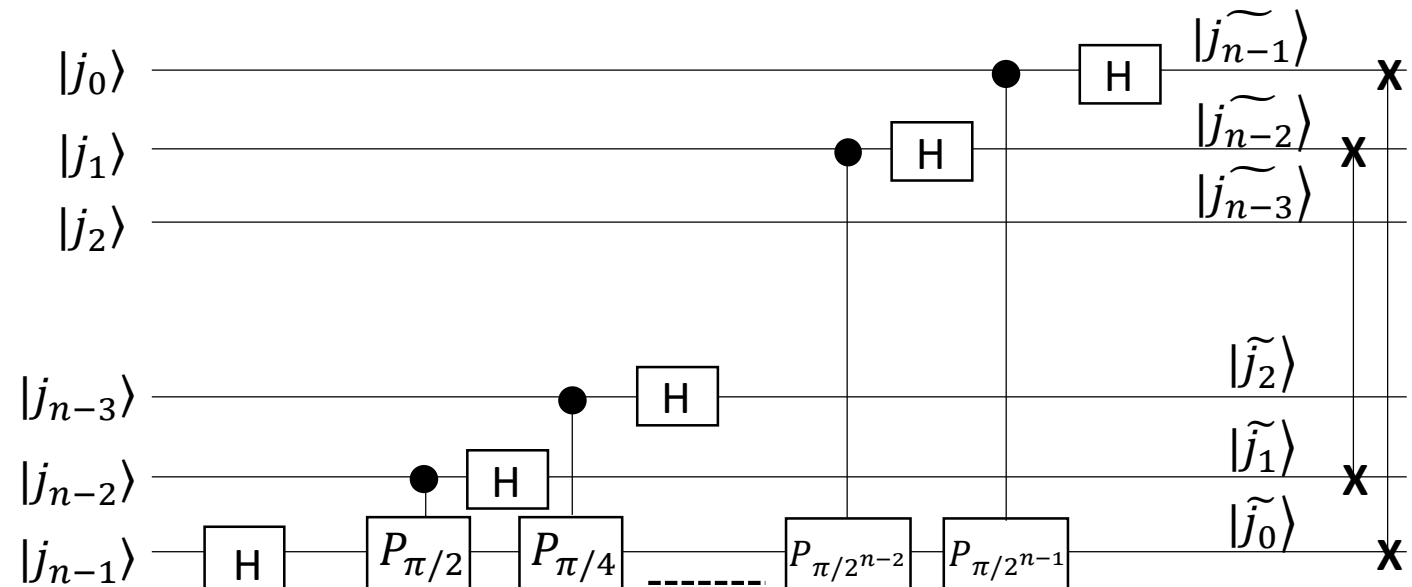
⋮

$$|\widetilde{j_1}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi j_{n-2}} e^{i\frac{\pi}{2} j_{n-3}} \dots e^{i\frac{\pi}{2^{n-3}} j_1} e^{i\frac{\pi}{2^{n-2}} j_0} |1\rangle)$$

$$|\widetilde{j_0}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi j_{n-1}} e^{i\frac{\pi}{2} j_{n-2}} \dots e^{i\frac{\pi}{2^{n-2}} j_1} e^{i\frac{\pi}{2^{n-1}} j_0} |1\rangle)$$

$$|j\rangle = |j_{n-1}\rangle \otimes |j_{n-2}\rangle \otimes |j_{n-3}\rangle \dots \otimes |j_1\rangle \otimes |j_0\rangle$$

QFT simply add phase  
to a quantum state





# n-Qubits QFT (Nielsen and Chuang)

With a little algebra the quantum Fourier transform can be given the following useful *product representation*:

$$|j_1, \dots, j_n\rangle \rightarrow \frac{\left(|0\rangle + e^{2\pi i 0 \cdot j_n} |1\rangle\right) \left(|0\rangle + e^{2\pi i 0 \cdot j_{n-1} j_n} |1\rangle\right) \dots \left(|0\rangle + e^{2\pi i 0 \cdot j_1 j_2 \dots j_n} |1\rangle\right)}{2^{n/2}}. \quad (5.4)$$

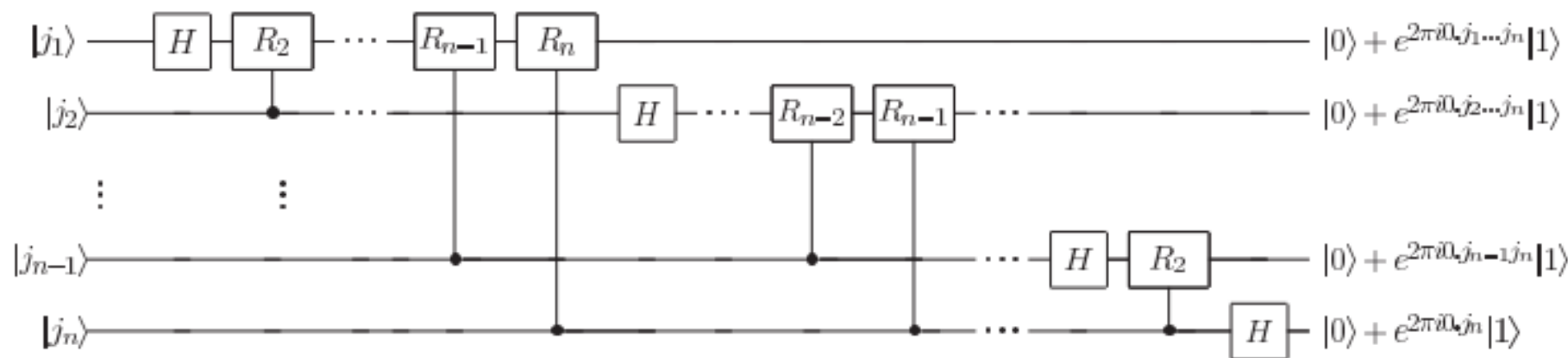
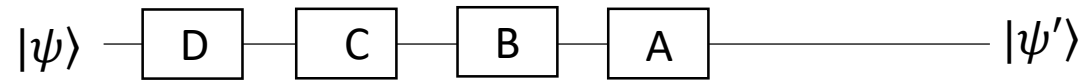


Figure 5.1. Efficient circuit for the quantum Fourier transform. This circuit is easily derived from the product representation (5.4) for the quantum Fourier transform. Not shown are swap gates at the end of the circuit which reverse the order of the qubits, or normalization factors of  $1/\sqrt{2}$  in the output.

# Inverse QFT

$$|\psi'\rangle = \hat{U} |\psi\rangle \quad \hat{U} = ABCD$$

$$|\psi'\rangle = ABCD|\psi\rangle$$

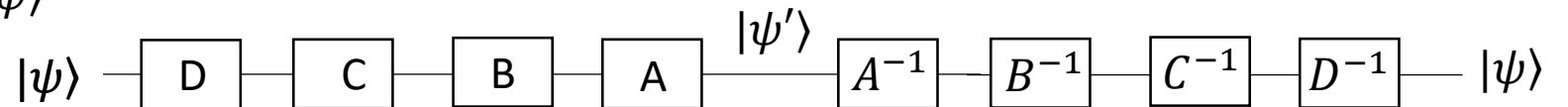


To get the initial state  $|\psi\rangle$  back, what gate sequence you need to apply

$$\hat{U}' = D^\dagger C^\dagger B^\dagger A^\dagger = D^{-1} C^{-1} B^{-1} A^{-1}$$

$$\hat{U}' |\psi'\rangle = (D^{-1} C^{-1} B^{-1} A^{-1}) ABCD |\psi\rangle$$

$$\hat{U}' |\psi'\rangle = |\psi\rangle$$



For inverse QFT, use these inverse operation in correct order

$$P_\theta \rightarrow P_{-\theta}$$

$$H \rightarrow H$$

# Period Finding and Prime Factorization

Consider a natural number  $N > 2$

How many qubits ( $n$ ) are required to represent the  $N$ ?

$$N = 13 \quad n = 4 \quad N = 2^n \Rightarrow N = 2^4 = 16$$

$$N = 15 \quad n = 4 \quad N = 2^n \Rightarrow N = 2^4 = 16$$

$$N = 21 \quad n = 5 \quad N = 2^n \Rightarrow N = 2^5 = 32$$

$$n = \lceil \log_2 N \rceil$$

$n$  – number of qubits

$N$  – natural number

$$2^n = N$$

Consider another natural number  $m$ , where

$$m = 2n$$

$$m = 2n = 2\lceil \log_2 N \rceil$$

$$m = \lceil \log_2 N^2 \rceil = \lceil \log_2 q \rceil$$

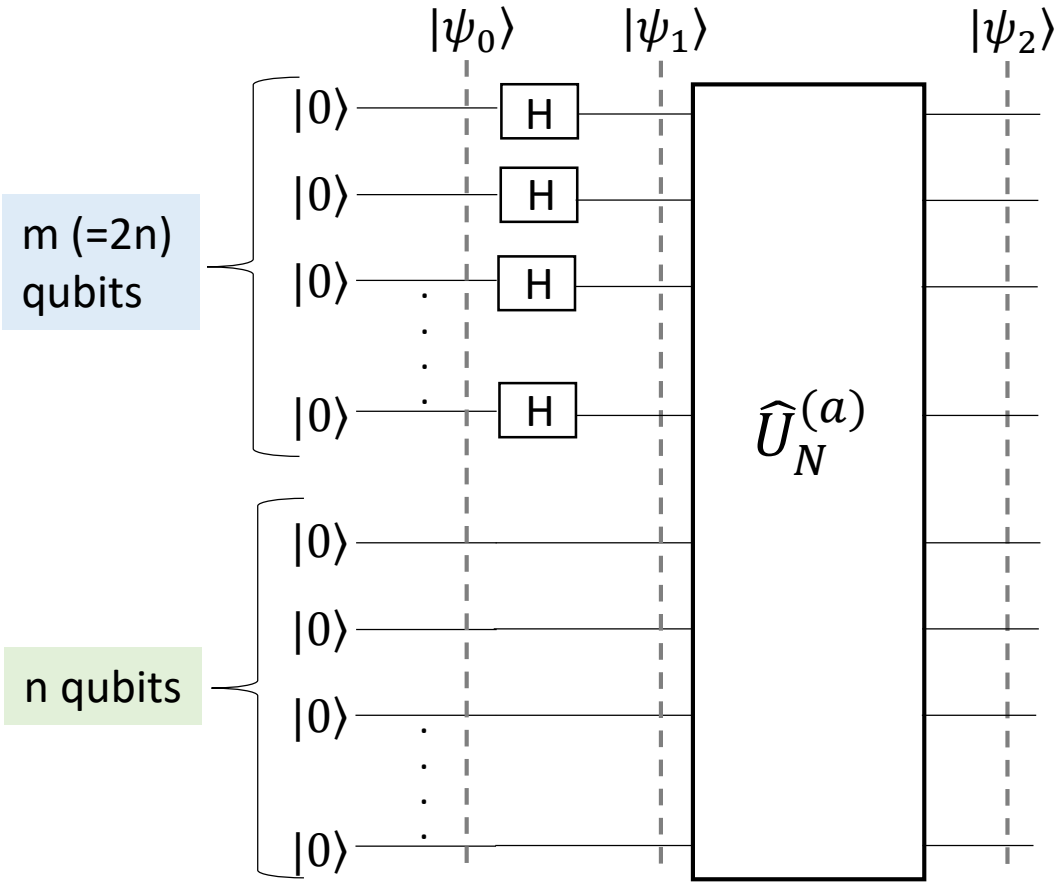
$$2^m = q$$

For  $N = 14$

$n = 4$  (qubits)  
16 combination of 0 and 1

$m = 8$  (qubits)  
256 combination of 0 and 1

# Period Finding and Prime Factorization



$$|\psi_0\rangle = |0\rangle^{\otimes m} \otimes |0\rangle^{\otimes n}$$

$$|\psi_1\rangle = \hat{H}^{\otimes m}(|0\rangle^{\otimes m}) \otimes |0\rangle^{\otimes n}$$

$$|\psi_1\rangle = \frac{1}{\sqrt{q}} \sum_{l=0}^{q-1} |l\rangle \otimes |0\rangle^{\otimes n}$$

$$2^m = q$$

$$\hat{U}_N^{(a)} |x\rangle \otimes |y\rangle = |x\rangle \otimes |y \oplus a^x \pmod{N}\rangle$$

Remember

$|0\rangle^{\otimes n}$  is string of zero whose integer/decimal value is always zero

$$|\psi_2\rangle = \frac{1}{\sqrt{q}} \sum_{l=0}^{q-1} |l\rangle \otimes |0 \oplus a^l \pmod{N}\rangle$$

$$|\psi_2\rangle = \frac{1}{\sqrt{q}} \sum_{l=0}^{q-1} |l\rangle \otimes |a^l \pmod{N}\rangle$$

# What is Modular Arithmetic

$$\frac{6}{5} \Rightarrow 6 \bmod 5 \Rightarrow 6 = 1 \pmod{5}$$

$$\frac{7}{5} \Rightarrow 7 \bmod 5 \Rightarrow 7 = 2 \pmod{5}$$

$$\frac{5^3}{44} \Rightarrow 5^3 \bmod 44 \Rightarrow 5^3 = 37 \pmod{44}$$

Now modular in terms of **a** and **N**

$$1 < a < N$$

$$\gcd(a, N) = 1$$

Let  $a = 2$  and  $N = 21$

$$a^0 = 1 \pmod{21}$$

$$a^{10} = 16 \pmod{21}$$

$$a^1 = 2 \pmod{21}$$

$$a^{11} = 11 \pmod{21}$$

$$a^2 = 4 \pmod{21}$$

$$a^{12} = 1 \pmod{21}$$

$$a^3 = 8 \pmod{21}$$

$$a^{13} = 2 \pmod{21}$$

$$a^4 = 16 \pmod{21}$$

$$a^{14} = 4 \pmod{21}$$

$$a^5 = 11 \pmod{21}$$

$$a^{15} = 8 \pmod{21}$$

$$a^6 = 1 \pmod{21}$$

$$a^7 = 2 \pmod{21}$$

$$a^8 = 4 \pmod{21}$$

$$a^9 = 8 \pmod{21}$$

Let  $a = 2$  and  $N = 9$

$$a^0 = 1 \pmod{9}$$

$$a^1 = 2 \pmod{9}$$

$$a^2 = 4 \pmod{9}$$

$$a^3 = 8 \pmod{9}$$

$$a^4 = 7 \pmod{9}$$

$$a^5 = 5 \pmod{9}$$

$$a^6 = 1 \pmod{9}$$

$$a^7 = 2 \pmod{9}$$

$$a^8 = 4 \pmod{9}$$

$$a^9 = 8 \pmod{9}$$

$$a^{10} = 7 \pmod{9}$$

$$a^{11} = 5 \pmod{9}$$

$$a^{12} = 1 \pmod{9}$$

$$a^{13} = 2 \pmod{9}$$

$$a^{14} = 4 \pmod{9}$$

$$a^{15} = 8 \pmod{9}$$

$f(x) = a^x$  is a periodic function

# What is Modular Arithmetic

$f(x) = a^x \pmod{N}$  is a periodic function

Smallest  $r > 0$  for which

$$a^r \pmod{N} = 1$$

$a^r$  is periodic with period  $r$

If  $r$  is even (an important property of periodic  $a^r$ )

$$a^r \pmod{N} = 1 \Rightarrow a^r = 1 \Rightarrow a^r - 1 = 0 \Rightarrow (a^{r/2} + 1)(a^{r/2} - 1) = 0$$

$$(a^{r/2} + 1)(a^{r/2} - 1) = \text{Multiple of } N$$



are factors of  $N$ , prime ? We don't know

# Period Finding and Prime Factorization

$$|\psi_2\rangle = \frac{1}{\sqrt{q}} \sum_{l=0}^{q-1} |l\rangle \otimes |a^l \pmod{N}\rangle$$

What is measurement outcome of  $|l\rangle$  and  $|a^l\rangle$

Now by expanding summation over 'l'

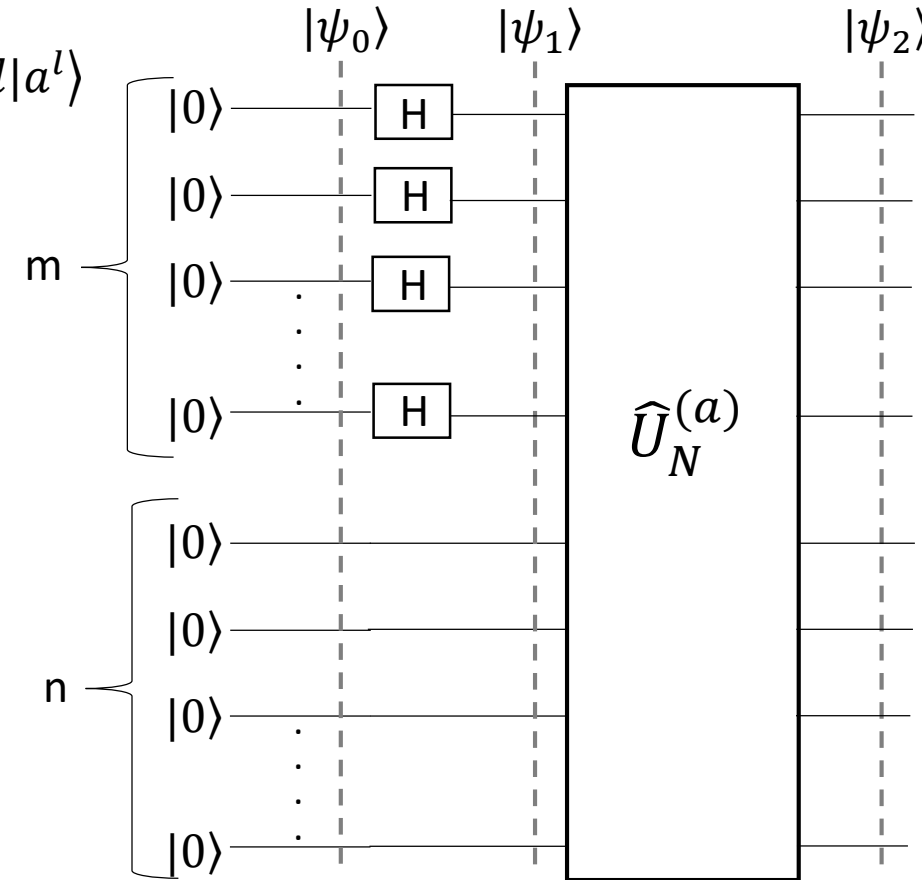
Assuming 'r' is period and we expect  $r < q$

$$\sqrt{q}|\psi_2\rangle = |0\rangle \otimes |1\rangle + |1\rangle \otimes |a^1\rangle + |2\rangle \otimes |a^2\rangle + \dots |r-1\rangle \otimes |a^{r-1}\rangle$$

$$+ |r\rangle \otimes |1\rangle + |r+1\rangle \otimes |a^1\rangle + |r+2\rangle \otimes |a^2\rangle + \dots |2r-1\rangle \otimes |a^{r-1}\rangle$$

$$+ |2r\rangle \otimes |1\rangle + |2r+1\rangle \otimes |a^1\rangle + |2r+2\rangle \otimes |a^2\rangle + \dots |3r-1\rangle \otimes |a^{r-1}\rangle$$

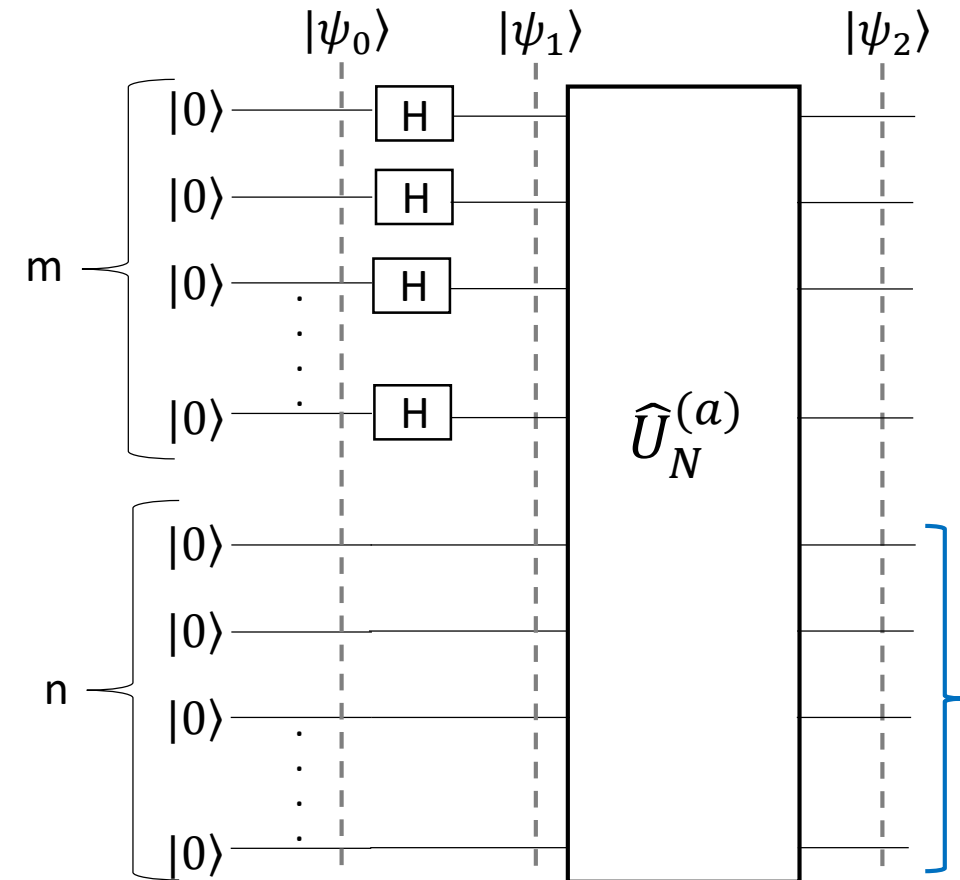
$$+ |(c-1)r\rangle \otimes |1\rangle + |(c-1)(r+1)\rangle \otimes |a^1\rangle + |(c-1)(r+2)\rangle \otimes |a^2\rangle + \dots |(c-1)(r+r_0)\rangle \otimes |a^{r-r_0}\rangle$$



The last term assuming 'r' is not xple of 'q', if it is, substitute  $r_0 = 0$

Number terms (strings of '0 and 1') in first register is very large compared to 2<sup>nd</sup> register

# Period Finding and Prime Factorization



$$|\psi_2\rangle = \frac{1}{\sqrt{q}} \sum_{l=0}^{q-1} |l\rangle \otimes |a^l \pmod{N}\rangle$$

What is measurement outcome of  $|l\rangle$  and  $|a^l\rangle$

What are the possible measurement outcomes of 2<sup>nd</sup> qubit string  $|a^l\rangle$

2<sup>nd</sup> qubit will collapse to  $Z = a^{r_1}$  Where  $0 \leq r_1 < r$

What is the probability of measurement this state? And upon measuring 2<sup>nd</sup> qubit (string) what is the state of the 1<sup>st</sup> qubit (string)

After measuring  $|\psi_2\rangle$  the new state will be

$$|\psi_3\rangle = \frac{1}{\sqrt{c}} \sum_{k=0}^{c-1} |kr + r_1\rangle \otimes |a^{r_1}\rangle$$

1<sup>st</sup> qubit (string) state

$$|\psi'_3\rangle = |r_1\rangle + |r + r_1\rangle + |2r + r_1\rangle \dots | (c - 1)r + r_1 \rangle$$

$$\text{Where } c = \left\lceil \frac{q}{r} \right\rceil \text{ or } \left\lfloor \frac{q}{r} \right\rfloor$$

Now leave the 2<sup>nd</sup> qubit (string) and consider only 1<sup>st</sup> qubit string



# Period Finding and Prime Factorization

Still we don't know what is period  $r$ . To find  $r$  we apply 'Inv QFT'

